

Linkage Disequilibrium Questions

1.) For a 1,000 chromosomes the following haplotypes were observed.

A_1B_1 200
 A_1B_2 50
 A_2B_1 350
 A_2B_2 400

a) What is the allele frequency for the A_1 allele and A_2 allele _____.

b.) What is the allele frequency for the B_1 and B_2 allele _____.

c.) What are the expected haplotype frequencies under linkage equilibrium

$P_{11} = A_1B_1$ _____

$P_{12} = A_1B_2$ _____

$P_{21} = A_2B_1$ _____

$P_{22} = A_2B_2$ _____

2.) Please answer the following for the above problem.

$D =$ _____

$D' =$ _____

$r^2 =$ _____

Is there statistical evidence that Marker A and B are in linkage disequilibrium _____?

$X^2 =$ _____ $p\text{-value} =$ _____